Supplementary Materials for
Predicting the long-term impact of AI assistants on collective intelligence

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Supplementary Text

Here, we derive the expected payoffs and collective accuracy used in the simulations. We consider Omniscient AI and Chatbot AI separately because they contain different amounts of information and therefore require different aggregation rules. We then analyze the corresponding evolutionary dynamics under a deterministic large-population approximation.

Omniscient AI

We first consider Omniscient AI, which incorporates all causal factors regardless of the player's interest. We derive individual and collective predictions before calculating the expected payoffs under each incentive structure.

Model

Because Omniscient AI observes all M causal factors, its prediction for player A is given by

$$Y_A^{\mathrm{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon, \tag{S1}$$

where b_i represent factor-specific biases and ϵ represents stochastic error, following a Gaussian distribution with mean zero and standard deviation τ. A new stochastic error is generated independently for each player and prediction event.

Let $B = \sum_{i=0}^{M} b_i$ denote the total additive bias. Because the true prediction target is $Y = \alpha_0 + \sum_{i=1}^{M} \alpha_i X_i$, the AI prediction can be written as

$$Y^{AI} = Y + B + \epsilon. \tag{S2}$$

In contrast to the AI prediction, player A's initial prediction depends only on the causal factor associated with their interest. Their initial and revised predictions are therefore given by:

$$Y_A = c_A X_{e_A} \tag{S3}$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \tag{S4}$$

represents player A's interest, c_A represents their belief about the effect of the corresponding factor, and β_A represents their reliance on AI. As in the main text, we set $X_0 = 1$, such that a player with $e_A = 0$ makes a constant initial prediction.

Aggregation

Because the AI component of each revised prediction contains information about all M causal factors, we use simple averaging to obtain the collective prediction:

$$\hat{Y}^{re} = \frac{1}{N}\sum_{j=1}^{N} Y_j^{re} \tag{S5}$$

$$= \frac{1}{N}\sum_{j=1}^{N} \left\{\beta_j Y^{AI} + (1-\beta_j)Y_j\right\} \tag{S6}$$

$$= \frac{1}{N}\sum_{j=1}^{N} \left\{\beta_j (Y + B + \epsilon_j) + (1-\beta_j)c_j X_{e_j}\right\} \tag{S7}$$

To separate the contributions of AI and the players' initial predictions, we define the population's mean reliance on AI, the stochastic AI error remaining after aggregation, and the aggregate contribution of initial predictions associated with each factor:

$$\bar{\beta} = \frac{1}{N}\sum_{j=1}^{N} \beta_j \tag{S8}$$

$$\bar{\eta} = \frac{1}{N}\sum_{j=1}^{N} \beta_j \epsilon_j \tag{S9}$$

$$\gamma_i = \frac{1}{N}\sum_{A \in \mathcal{A}_i} (1-\beta_A)c_A \tag{S10}$$

Here, $\bar{\beta}$ represents the mean reliance on AI, $\bar{\eta}$ represents the stochastic AI error remaining after averaging, and γ_i represents the contribution of players' initial predictions to factor i. Using these definitions, the revised collective prediction can be written as follows:

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^{M} \gamma_i X_i + \bar{\eta} \tag{S11}$$

Thus, the collective prediction consists of an AI-derived component, the contributions of the players' initial predictions, and the stochastic AI error that remains after aggregation.

Expected payoffs

We next derive the expected payoff under each incentive structure. Expectations are taken over the distributions of the causal factors and stochastic AI errors, conditional on the current population strategies and the fixed parameters of the prediction target.

Feedback

Under the feedback structure, a player's payoff depends on the alignment between their revised prediction and the residual error of the collective prediction. To express this residual, we first define the remaining error in the intercept and in the coefficient of each causal factor:

$$\delta_0 = \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0), \tag{S12}$$

$$\delta_i = \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \qquad (i \neq 0), \tag{S13}$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^{M} \delta_i X_i - \bar{\eta}, \tag{S14}$$

where δ_0 represents the remaining error in the intercept, whereas δ_i represents the remaining error associated with factor i. Because the causal factors and stochastic AI errors are independent and have zero mean, cross terms vanish when the expectation is taken. Substituting the revised individual prediction and collective residual into the feedback payoff gives

$$\mathbb{E}[\pi_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \tag{S15}$$

$$= \mathbb{E}\left[\left\{\beta_A\alpha_0 + \beta_A \sum_{i=1}^{M} \alpha_i X_i + \beta_A B + \beta_A \epsilon) + (1-\beta_A)c_A X_{e_A}\right\}\right. \tag{S16}$$

$$\left. \times \left\{\delta_0 + \sum_{i=1}^{M} \delta_i X_i - \bar{\eta}\right\}\right] \tag{S17}$$

$$= \beta_A\alpha_0\delta_0 + \beta_A \sum_{i=1}^{M} \delta_i\alpha_i\sigma_i^2 + \beta_A B\delta_0 - \frac{1}{N}\beta_A^2\tau^2 + (1-\beta_A)c_A\delta_{e_A}\sigma_{e_A}^2 \tag{S18}$$

$$= \beta_A \left\{(\alpha_0 + B)\delta_0 + \sum_{i=1}^{M} \alpha_i\delta_i\sigma_i^2\right\} + (1-\beta_A)c_A\delta_{e_A}\sigma_{e_A}^2 - \frac{\beta_A^2\tau^2}{N} \tag{S19}$$

The first term represents the alignment between the AI-derived prediction and the collective residual, whereas the second term represents the contribution of the player's initial prediction. The final term arises because the stochastic error in player A's AI prediction also contributes to the collective prediction. This finite-population effect decreases as population size increases.

Niche-expert

Under the niche-expert structure, the payoff is the negative squared error of the revised individual prediction, weighted by the frequency of the player's interest. Because $X_0 = 1$, a player with $e_A = 0$ makes a constant initial prediction. In this

case, the expected payoff is:

$$\mathbb{E}[\pi_A] = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \tag{S20}$$

$$= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\right\}^2\right] \tag{S21}$$

$$= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^{M}\alpha_i X_i + \beta_A \epsilon\right\}^2\right] \tag{S22}$$

$$= -\rho_0\Big\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2 \tag{S23}$$

$$+ (1 - \beta_A)^2\sum_{i\neq 0}\alpha_i^2\sigma_i^2 + \beta_A^2\tau^2\Big\} \tag{S24}$$

For a player with $e_A \neq 0$, the initial prediction instead depends on the observed value of factor X_{e_A}. The expected payoff is therefore:

$$\mathbb{E}[\pi_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \tag{S25}$$

$$= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\right\}^2\right] \tag{S26}$$

$$= -\mathbb{E}\Bigg[\rho_{e_A}\Big\{(\beta_A - 1)\alpha_0 + \beta_A B \tag{S27}$$

$$+ (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i\neq e_A,0}\alpha_i X_i + \beta_A \epsilon\Big\}^2\Bigg] \tag{S28}$$

$$= -\rho_{e_A}\Big\{(1 - \beta_A)^2\alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B \tag{S29}$$

$$+ \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2\sigma_{e_A}^2 \tag{S30}$$

$$+ (1 - \beta_A)^2\sum_{i\neq e_A,0}\alpha_i^2\sigma_i^2 + \beta_A^2\tau^2\Big\} \tag{S31}$$

These expressions separate the contributions of the player's belief, the unobserved causal factors, systematic AI bias, and stochastic AI error. Increasing reliance on AI reduces the contribution of errors in the player's initial prediction but increases their exposure to AI bias and stochastic error.

Incentivizing or penalizing AI use

Direct incentives add the term $\lambda\beta_A$ to either the feedback or niche-expert payoff. Because this term is fixed conditional on the player's strategy, its expectation is simply $\lambda\beta_A$. The expected payoffs therefore become

$$\mathbb{E}[\pi_{A,\text{Feedback}}] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A, \tag{S32}$$

$$\mathbb{E}[\pi_{A,\text{Niche-expert}}] = \mathbb{E}\left[-\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \tag{S33}$$

Thus, a positive value of λ increases the payoff of strategies with greater reliance on AI, whereas a negative value favors strategies with lower reliance.

Expected accuracy

Using the residual defined above, the mean squared error of the revised collective prediction is:

$$\mathbb{E}[(Y-\alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^{M}\alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^{M}\alpha_i^2\sigma_i^2 \tag{S34}$$

$$\mathbb{E}[(Y-\hat{Y}^{re})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^{M}\delta_i X_i - \bar{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^{M}\delta_i^2\sigma_i^2 + \frac{\tau^2}{N^2}\sum_{j=1}^{N}\beta_j^2 \tag{S35}$$

The expected collective accuracy is therefore

$$R = 1 - \frac{\mathbb{E}[(Y-\hat{Y}^{re})^2]}{\mathbb{E}[(Y-\alpha_0)^2]} \tag{S36}$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^{M}\delta_i^2\sigma_i^2 + \frac{\tau^2}{N^2}\sum_{j=1}^{N}\beta_j^2}{\sum_{i=1}^{M}\alpha_i^2\sigma_i^2} \tag{S37}$$

The collective prediction error consists of the squared error in the intercept, the remaining error associated with each causal factor, and the stochastic AI error remaining after averaging across players. Counterfactual collective accuracy is obtained from the same expression by setting $\beta_A = 0$ for all players while retaining their evolved interests and beliefs.

Chatbot AI

Unlike Omniscient AI, Chatbot AI use only the causal factor associated with each player's interest. This causes the information contained in AI predictions to depend directly on the distribution of interests in the population. Here, we derive the collective prediction, expected payoffs, and expected collective accuracy under this form of AI assistance.

Model

The prediction generated by Chatbot AI for player A depends on the causal factor associated with the player's interest:

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon, \tag{S38}$$

where b_{e_A} represents factor-specific additive bias, and ϵ represents stochastic error. The stochastic error follows a Gaussian distribution with mean zero and standard

deviation τ and is generated independently across players and prediction events. Player A's initial and revised predictions can be then written as follows:

$$Y_A = c_A X_{e_A} \tag{S39}$$

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \tag{S40}$$

Here, c_A represents player A's belief about the effect of their focal factor, e_A represents their interest, and β_A represents their reliance on AI. Thus, the revised prediction remains specific to the single factor associated with the player's interest.

Aggregation

Because both the player's initial prediction and the Chatbot AI prediction contain information about the same causal factor, we use clustering aggregation. Revised predictions are first averaged among players interested in the same factor and then summed across represented interests:

$$\hat{Y}^{re} = \sum_{i=0}^{M} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \tag{S41}$$

$$= \sum_{i=0}^{M} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \left\{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \right\} \tag{S42}$$

$$= \sum_{i=0}^{M} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \left[\left\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \right\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon_A \right], \tag{S43}$$

where $\mathcal{A}_i$ is the set of players interested in factor i, and $n_i = |\mathcal{A}_i|$ is the number of players in that set.

To simplify the collective prediction, we define the effective coefficient, weighted AI bias, and stochastic error associated with each interest group:

$$\tilde{c}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \left\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \right\}, \tag{S44}$$

$$\tilde{B}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A}, \tag{S45}$$

$$\bar{B} = \sum_{i=0}^{M} \tilde{B}_i, \tag{S46}$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon_A, \tag{S47}$$

$$\bar{\eta} = \sum_{i=0}^{M} \tilde{\eta}_i, \tag{S48}$$

where $\tilde{c}_i$ represents the mean effective coefficient for factor i after AI assistance, whereas $\tilde{B}_i$ and $\tilde{\eta}_i$ represent its systematic and stochastic error components. Thus, the collective prediction can be written as:

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \tilde{c}_i X_i + \bar{B} + \bar{\eta}, \tag{S49}$$

where $\mathcal{I} = \{i \in \{0, \ldots, M\} | n_i > 0\}$ denotes the set of represented interests.

Expected payoffs

We next derive the expected payoff under each incentive structure. Expectations are taken over the causal factors and stochastic AI errors, conditional on the current population strategies and the fixed parameters of the prediction target.

Feedback

Under the feedback structure, players are rewarded according to whether their revised predictions reduce the error in the collective prediction. We therefore define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - \tilde{c}_0 - \bar{B}, \tag{S50}$$

$$\delta_i = \alpha_i - \tilde{c}_i, \qquad (i \neq 0), \tag{S51}$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^{M} \delta_i X_i - \bar{\eta}, \tag{S52}$$

where δ_0 includes the remaining error in the intercept and the total additive AI bias, whereas δ_i measures the remaining error in the coefficient of factor i. Because the causal factors and stochastic errors are independent and have zero mean, substituting the revised individual prediction into the feedback payoff gives

$$\mathbb{E}[\pi_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \tag{S53}$$

$$= \mathbb{E}\left[\left\{\left(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\right) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\right\}\right. \tag{S54}$$

$$\left. \times \left\{ \delta_0 + \sum_{i=1}^{M} \delta_i X_i - \bar{\eta} \right\} \right] \tag{S55}$$

$$= \left\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\right\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \tag{S56}$$

Thus, the expected feedback payoff depends on the remaining collective error associated with the player's interest, the contribution of factor-specific AI bias to the intercept error, and the player's contribution to the stochastic error of their interest group.

Niche-expert

Under the niche-expert structure, the negative squared error of each revised individual prediction is weighted by the frequency of the player's interest. We consider $e_A = 0$ separately because these players make constant predictions rather than factor-dependent predictions. When $e_A = 0$, the expected payoff is

$$\mathbb{E}[\pi_A] = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \tag{S57}$$

$$= -\mathbb{E}\left[\rho_0\left[\{\beta_A\alpha_0 + (1-\beta_A)c_A\} + \beta_A b_0 + \beta_A\epsilon - \alpha_0 - \sum_{i=1}^{M}\alpha_i X_i\right]^2\right] \tag{S58}$$

$$= -\rho_0\Big\{(1-\beta_A)^2(c_A-\alpha_0)^2 + 2(1-\beta_A)(c_A-\alpha_0)\beta_A b_0 + \beta_A^2 b_0^2 \tag{S59}$$

$$+ \sum_{i\neq 0}\alpha_i^2\sigma_i^2 + \beta_A^2\tau^2\Big\} \tag{S60}$$

When $e_A \neq 0$, the player's prediction varies with their focal causal factor. The expected payoff is then

$$\mathbb{E}[\pi_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \tag{S61}$$

$$= -\mathbb{E}\left[\rho_{e_A}\left[\{\beta_A\alpha_{e_A} + (1-\beta_A)c_A\}X_{e_A}\right.\right. \tag{S62}$$

$$\left.\left. + \beta_A b_{e_A} + \beta_A\epsilon - \alpha_0 - \sum_{i=1}^{M}\alpha_i X_i\right]^2\right] \tag{S63}$$

$$= -\rho_{e_A}\Big\{(1-\beta_A)^2(c_A-\alpha_{e_A})^2\sigma_{e_A}^2 + \sum_{i\neq e_A,0}\alpha_i^2\sigma_i^2 \tag{S64}$$

$$+ \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0\beta_A b_{e_A} + \beta_A^2\tau^2\Big\} \tag{S65}$$

In both cases, the expected squared error consists of the mismatch between the player's belief and the true coefficient, contributions from causal factors omitted from the individual prediction, factor-specific AI bias, and stochastic AI error. The niche-expert weight ρ_{e_A} scales these errors according to the frequency of the player's interest.

Incentivizing or penalizing AI use

Directly incentivizing or penalizing AI use adds the term $\lambda\beta_A$ to the player's payoff. Because this term is fixed conditional on the player's strategy, it can be taken outside

the expectation. The expected payoffs are therefore

$$\mathbb{E}[\pi_{A,Feedback}] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] \tag{S66}$$

$$= \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A, \tag{S67}$$

$$\mathbb{E}[\pi_{A,Niche-expert}] = \mathbb{E}\left[-\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] \tag{S68}$$

$$= -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \tag{S69}$$

Balanced

Under the balanced structure, the payoff corresponds to the weighted mean of the feedback and niche-expert payoff components, with balancing weight w:

$$\mathbb{E}[\pi_A] = \mathbb{E}\left[(1-w)Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - w\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] \tag{S70}$$

$$= (1-w)\mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] - w\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] \tag{S71}$$

The expected payoff under the balanced structure can therefore be calculated directly from the feedback and niche-expert payoffs derived above.

Unrevised balanced

The unrevised feedback and niche-expert components are identical to those in the model without AI assistance (Wang et al., 2025). We therefore calculate their expected payoffs following the derivations in the original model and take their arithmetic mean:

$$\mathbb{E}[\pi_A] = \frac{1}{2}\mathbb{E}\left[Y_A(Y - \hat{Y}) - \rho_{e_A}(Y_A - Y)^2\right]. \tag{S72}$$

Combined balanced

Under the combined balanced structure, the payoff is the arithmetic mean of the balanced and unrevised balanced payoffs, with $w = 0.5$ for both balanced components. Its expected payoff is therefore

$$\mathbb{E}[\pi_A] = \frac{1}{4}\mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2\right. \tag{S73}$$

$$\left. + Y_A(Y - \hat{Y}) - \rho_{e_A}(Y_A - Y)^2\right]. \tag{S74}$$

The expected payoff under the combined balanced structure can therefore be calculated directly from the revised and unrevised feedback and niche-expert payoffs derived above.

Expected accuracy

Finally, we derive the expected collective accuracy under clustering aggregation. The intercept α_0 serves as the baseline prediction. Because the causal factors are independent and have mean zero, its expected squared error is

$$\mathbb{E}[(Y-\alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^{M} \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^{M} \alpha_i^2 \sigma_i^2 \tag{S75}$$

Using the expression for the collective residual above, the expected squared error of the revised collective prediction is

$$\mathbb{E}[(Y-\hat{Y}^{\mathrm{re}})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^{M} \delta_i X_i - \bar{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^{M} \delta_i^2 \sigma_i^2 + \sum_{i=0}^{M} \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \tag{S76}$$

The first two terms represent squared errors in the intercept and factor-specific coefficients. The final term represents the stochastic AI error remaining after predictions are averaged within each interest group. Thus, stochastic error has a larger effect when an interest is represented by fewer individuals. Normalizing this error relative to the baseline gives:

$$R = 1 - \frac{\mathbb{E}[(Y-\hat{Y}^{re})^2]}{\mathbb{E}[(Y-\alpha_0)^2]} \tag{S77}$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^{M} \delta_i^2 \sigma_i^2 + \sum_{i=0}^{M} \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^{M} \alpha_i^2 \sigma_i^2} \tag{S78}$$

Here, $R = 1$ represents perfect collective prediction, $R = 0$ represents performance equal to the baseline, and $R < 0$ represents performance worse than the baseline.

Evolutionary dynamics

Following previous work (Wang et al., 2025; Taylor and Jonker, 1978; Traulsen et al., 2005), we use replicator dynamics as a deterministic large-population approximation to characterize equilibrium states and their local stability. The agent-based simulations themselves follow the Fermi imitation rule defined above. At equal-payoff stationary states, the replicator and Fermi formulations imply the same local invasion directions for any finite selection strength $s > 0$, although their transient dynamics need not coincide. We focus on the feedback and niche-expert structures, for which the long-term dynamics can be characterized analytically. The direct interventions on AI use and the balanced structure are examined through simulations.

Let λ denote a strategy, p_λ its frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff. Then, the corresponding replicator equation can be written as:

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \tag{S79}$$

where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$ represents the mean expected payoff of the population. At an internal stationary state, all strategies with positive frequency must have the same expected payoff. A boundary state can also be stationary because a strategy with zero frequency cannot increase unless it is introduced into the population. We therefore assess the stability of boundary states by asking whether they can be invaded by a rare alternative strategy.

Omniscient AI

With Omniscient AI, every AI prediction contains information about the full prediction target, and revised predictions are combined through averaging. As population size increases, independent AI errors vanish from the collective prediction. However, these errors remain in individual prediction errors and can therefore continue to affect niche-expert payoffs. We consider the two incentive structures separately.

Feedback

Under the feedback structure, both the stochastic error in the collective prediction, $\bar{\eta}$, and its contribution to expected payoff vanish as $N \to \infty$. Following previous work (Wang et al., 2025), we define the Lyapunov function as the expected error of collective prediction:

$$V = \mathbb{E}\left[(Y - \hat{Y}^{re})^2\right]. \tag{S80}$$

The function V is non-negative and equals zero when the collective prediction is identical to the target. To determine how V changes over time, we take its time derivative:

$$\frac{dV}{dt} = -2\mathbb{E}\left[(Y - \hat{Y}^{re})\frac{d\hat{Y}^{re}}{dt}\right] \tag{S81}$$

$$= -2\mathbb{E}\left[(\delta_0 + \sum_{i=1}^{M}\delta_i X_i)\left\{\frac{d\bar{\beta}}{dt}(Y + B) + \sum_{i=0}^{M}\frac{d\gamma_i}{dt}X_i\right\}\right] \tag{S82}$$

$$= -2\left[\left\{(\alpha_0 + B)\delta_0 + \sum_{i=1}^{M}\alpha_i\delta_i\sigma_i^2\right\}\frac{d\bar{\beta}}{dt} + \delta_0\frac{d\gamma_0}{dt} + \sum_{i=1}^{M}\delta_i\sigma_i^2\frac{d\gamma_i}{dt}\right]. \tag{S83}$$

For notational convenience, we define

$$K = (\alpha_0 + B)\delta_0 + \sum_{i=1}^{M}\alpha_i\delta_i\sigma_i^2. \tag{S84}$$

Although K changes with the population state, it is common to all strategies at a given state and can therefore be treated as a constant with respect to the strategy index λ. We also define:

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \tag{S85}$$

Under the replicator equation, the rate of change in the population mean of a strategy component is given by its covariance with expected payoff.

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_\lambda (u_\lambda - \bar{u}) \beta_\lambda = \mathrm{Cov}(\beta, u). \tag{S86}$$

Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^{M} \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \mathrm{Cov}((1-\beta) c G_e, u). \tag{S87}$$

Therefore,

$$\frac{dV}{dt} = -2 \left\{ K \mathrm{Cov}(\beta, u) + \mathrm{Cov}((1-\beta) c G_e, u) \right\} \tag{S88}$$

$$= -2 \mathrm{Cov}(K\beta + (1-\beta) c G_e, u). \tag{S89}$$

For the feedback structure in the limit $N \to \infty$, the expected payoff of strategy λ is

$$u_\lambda = K\beta_\lambda + (1 - \beta_\lambda) c_\lambda G_{e_\lambda}. \tag{S90}$$

This expected payoff is identical to the quantity inside the covariance above. Therefore,

$$\frac{dV}{dt} = -2 \mathrm{Cov}(u, u) = -2 Var(u) \leq 0. \tag{S91}$$

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\mathrm{Var}(u) = 0$. This condition arises when all represented strategies receive the same expected payoff. However, this condition does not necessarily imply that $V = 0$, because a population can lack strategies that would improve its collective prediction.

If $V > 0$, at least one component of the collective residual remains nonzero. A rare strategy whose prediction is aligned with this residual can obtain a higher expected payoff than the resident strategies and invade the population. Thus, provided that novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$. At such a state, $\hat{Y}^{\mathrm{re}} = Y$, and collective accuracy reaches $R = 1$.

Niche-expert

Unlike feedback payoffs, niche-expert payoffs depend on the prediction error of each individual. Stochastic AI error therefore remains in the expected payoff even when $N \to \infty$, and we retain $\tau > 0$ throughout this analysis.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β. For a nonzero interest e, the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \tag{S92}$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β. Its unconstrained stationary point is then:

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e,0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e,0} \alpha_j^2 \sigma_j^2 + \tau^2}. \tag{S93}$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e, so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

To compare the optimized strategies across interests, let L_e denote the minimum expected individual squared error within interest e, $\mathbb{E}[(Y_A^{re} - Y)^2]$. For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e,0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e,0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e,0} \alpha_j^2 \sigma_j^2 + \tau^2}. \tag{S94}$$

If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating the expected squared prediction error at the corresponding boundary.

Interest 0 must be considered separately because $X_0 = 1$ and the player's belief determines a constant prediction. Optimizing with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \tag{S95}$$

Substituting this into the expected payoff gives a strictly concave function of β, with the unique optimum:

$$\beta_0^* = \frac{\sum_{j=1}^{M} \alpha_j^2 \sigma_j^2}{\sum_{j=1}^{M} \alpha_j^2 \sigma_j^2 + \tau^2}. \tag{S96}$$

Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^{M} \alpha_j^2 \sigma_j^2, \tag{S97}$$

and

$$L_0 = \frac{\tau^2 \sum_{j=1}^{M} \alpha_j^2 \sigma_j^2}{\sum_{j=1}^{M} \alpha_j^2 \sigma_j^2 + \tau^2}. \tag{S98}$$

At this optimum, the player adjusts their belief to offset the systematic AI bias. The stochastic component of AI error cannot be removed in the same way and therefore limits the benefit of AI reliance.

We denote the optimal strategy within interest e by S_e. When $\beta_e^* = 1$, different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class. A stable state can contain only the optimal strategy class within each represented interest because a suboptimal strategy can be invaded by S_e.

We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote the frequency of S_e. Its expected payoff is

$$u_e = -\rho_e L_e. \tag{S99}$$

Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator equation at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \tag{S100}$$

Together with $\sum_{e=0}^{M} \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \qquad D = \left(\sum_{j=0}^{M} \frac{1}{L_j} \right)^{-1}. \tag{S101}$$

This internal stationary state is locally stable. At equilibrium, all represented interests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one interest increases slightly above its equilibrium value, its payoff decreases because the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of an interest decreases below its equilibrium value, its payoff increases. The replicator dynamics therefore generate a restoring force that reduces small deviations in interest frequencies and drives the population back toward the internal stationary state.

If one or more interests are absent, the resulting boundary state can be stationary under the replicator equation, but it is unstable against invasion. The payoff of a rare absent interest approaches zero as $\rho_e \to 0$, whereas the mean payoff of the resident population is negative when $L_e > 0$, so the rare interest increases when introduced at sufficiently small frequency.

Thus, there exists a unique stable full-support equilibrium in terms of the optimal prediction strategies and their frequencies. Whether the constrained optimum has $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because these cases differ only through the resulting value of L_e. When $\beta_e^* = 1$, the strategy is selectively neutral with respect to c_e, but its prediction, payoff, and equilibrium interest frequency remain uniquely determined.

Having characterized this equilibrium, we next compare its collective prediction with a prediction made entirely by AI. In particular, the expected squared error of an Omniscient AI prediction is

$$\mathbb{E}\left[(Y - Y^{\mathrm{AI}})^2\right] = B^2 + \tau^2. \tag{S102}$$

For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. Therefore, because L_e is the minimum individual expected squared prediction error within interest e,

$$L_e \leq B^2 + \tau^2. \tag{S103}$$

Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^{M} \alpha_j^2 \sigma_j^2}{\sum_{j=1}^{M} \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \tag{S104}$$

Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^{M} \rho_e^* L_e < B^2 + \tau^2. \tag{S105}$$

The collective prediction is the average of individual revised predictions. By convexity of the squared error,

$$\mathbb{E}\left[(Y - \hat{Y}^{re})^2\right] \leq \sum_{e=0}^{M} \rho_e^* L_e. \tag{S106}$$

Therefore,

$$\mathbb{E}\left[(Y - \hat{Y}^{re})^2\right] < \mathbb{E}\left[(Y - Y^{\mathrm{AI}})^2\right]. \tag{S107}$$

Thus, the diversity of optimized predictions allows the population to outperform an individual AI prediction at equilibrium.

However, outperforming AI does not imply perfect collective accuracy. Under averaging, the non-AI contribution associated with a particular factor is provided only by individuals interested in that factor. The predictions of individuals with other interests therefore attenuate that factor's coefficient in the collective prediction. For a nonzero factor i, the residual coefficient of the stationary collective prediction can be written as

$$\delta_i^* = \alpha_i \left\{1 - \bar{\beta}^* - \rho_i^*(1 - \beta_i^*)\right\} = \alpha_i \sum_{e \neq i} \rho_e^*(1 - \beta_e^*). \tag{S108}$$

At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \tag{S109}$$

Because $\sum_{i=1}^{M} \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error. Hence,

$$\mathbb{E}\left[(Y - \hat{Y}^{re})^2\right] > 0, \tag{S110}$$

and consequently the stationary collective accuracy satisfies $R^* < 1$. Thus, the niche-expert structure preserves interest diversity and allows the population to outperform AI, but averaging prevents the population from reaching perfect collective accuracy.

AI models under alternative aggregation rules

In the main analysis, we assume that collective predictions under the omniscient AI and chatbot AI models are generated using averaging and clustering aggregation, respectively. However, the aggregation rules can, in principle, be swapped between the two AI models. To examine the consequences of these alternative combinations, we investigate their behavior through mathematical derivation and simulation.

Omniscient AI with clustering aggregation

Under clustering aggregation, the collective prediction based on individual predictions revised by the omniscient AI can be written as

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \tag{S111}$$

$$= \sum_{i \in \mathcal{I}} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \left\{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \right\} \tag{S112}$$

$$= \sum_{i \in \mathcal{I}} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \left[\beta_A (Y + B + \epsilon_A) + (1 - \beta_A) c_A X_{e_A} \right], \tag{S113}$$

where $\mathcal{A}_i$ is the set of players interested in factor i, $n_i = |\mathcal{A}_i|$ is the number of players in that set, and $\mathcal{I} = \{i \in \{0, \ldots, M\} : n_i > 0\}$ is the set of represented interests.

To simplify the collective prediction, we define:

$$c_i^{agg} = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A, \tag{S114}$$

$$\bar{\beta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A, \tag{S115}$$

$$\beta^{agg} = \sum_{i \in \mathcal{I}} \bar{\beta}_i, \tag{S116}$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon_A, \tag{S117}$$

$$\bar{\eta} = \sum_{i \in \mathcal{I}} \tilde{\eta}_i. \tag{S118}$$

Thus, the collective prediction can be written as

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} c_i^{agg} X_i + \beta^{agg}(Y + B) + \bar{\eta}. \tag{S119}$$

Because individual reliance on AI, β_A, is initialized independently from $U(0, 1)$, the expected initial value of β^{agg} is $|\mathcal{I}|/2$. When all $M + 1$ interests are represented, this becomes $(M+1)/2$. Thus, clustering aggregation incorporates the omniscient AI prediction repeatedly across interest groups, scaling its contribution to the collective prediction by a factor of approximately $(M+1)/2$ at initialization. For a large number of factors, such as $M = 50$, this produces an unrealistically large contribution of the AI prediction to the collective prediction. Therefore, clustering aggregation does not provide a reasonable aggregation rule for the omniscient AI model in the context of our study, as it repeatedly incorporates predictions of the entire outcome across interest groups.

Chatbot AI with averaging aggregation

Each prediction generated by Chatbot AI contains information about only one factor. Averaging these predictions does not repeatedly incorporate the full prediction target, but instead weights the contribution of each factor by the proportion of individuals interested in that factor. Under averaging aggregation, the collective prediction can be written as

$$\hat{Y}^{re} = \frac{1}{N} \sum_{A=1}^{N} Y_A^{re} \tag{S120}$$

$$= \frac{1}{N} \sum_{A=1}^{N} \left[\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon_A \right]. \tag{S121}$$

We define the aggregate revised coefficient associated with each factor, the aggregate AI bias, and the stochastic error as

$$c_i^{av} = \frac{1}{N} \sum_{A \in \mathcal{A}_i} \left\{ \beta_A \alpha_i + (1 - \beta_A) c_A \right\}, \tag{S122}$$

$$B^{av} = \frac{1}{N} \sum_{A=1}^{N} \beta_A b_{e_A}, \tag{S123}$$

$$\bar{\eta} = \frac{1}{N} \sum_{A=1}^{N} \beta_A \epsilon_A. \tag{S124}$$

Thus, the collective prediction becomes

$$\hat{Y}^{re} = \sum_{i=0}^{M} c_i^{av} X_i + B^{av} + \bar{\eta}. \tag{S125}$$

We define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - c_0^{av} - B^{av}, \tag{S126}$$

$$\delta_i = \alpha_i - c_i^{av}, \qquad (i \neq 0). \tag{S127}$$

Expected payoffs

Under the feedback structure, the expected payoff is

$$\mathbb{E}[\pi_A] = \{\beta_A \alpha_{e_A} + (1-\beta_A)c_A\}\,\delta_{e_A}\sigma_{e_A}^2 + \beta_A b_{e_A}\delta_0 - \frac{\beta_A^2 \tau^2}{N}, \tag{S128}$$

where $\sigma_0^2 = 1$. Because the niche-expert payoff depends only on individual prediction error and the frequency of the player's interest, its expected payoff is unaffected by the aggregation rule. It is therefore identical to the expected payoff derived for Chatbot AI with clustering aggregation.

Expected accuracy

The expected squared error of the collective prediction is

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \delta_0^2 + \sum_{i=1}^{M} \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{A=1}^{N} \beta_A^2. \tag{S129}$$

Therefore, the expected collective accuracy is

$$R = 1 - \frac{\delta_0^2 + \sum_{i=1}^{M} \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{A=1}^{N} \beta_A^2}{\sum_{i=1}^{M} \alpha_i^2 \sigma_i^2}. \tag{S130}$$

Simulation results

The effect of averaging depends on the incentive structure (Fig. S10). Under the feedback structure, social learning can compensate for the attenuation introduced by averaging by favoring factor-specific predictions that reduce the remaining collective error. In particular, the mean revised coefficient among individuals interested in factor i can approach α_i/ρ_i, such that its aggregate contribution approaches α_i. Consistent with this mechanism, collective accuracy reaches a near-perfect level across AI bias values. Collective bias and variance also remain near zero, although counterfactual human CI remains below one.

In contrast, the niche-expert structure rewards individual predictive accuracy rather than contributions to the collective prediction. Revised factor-specific coefficients therefore remain close to their corresponding true coefficients, such that $\bar{c}_i^{\mathrm{re}} \approx \alpha_i$. Averaging then reduces their aggregate contribution to $c_i^{\mathrm{av}} \approx \rho_i \alpha_i$. When interests are approximately uniformly distributed, $\rho_i \approx 1/(M+1)$. The attenuation of the nonzero factor coefficients alone then gives $R \approx 1 - M/(M+1)^2 = 0.039$

when $M = 50$, before accounting for the remaining intercept and stochastic errors. Accordingly, collective accuracy remains near zero even though interest diversity remains high and reliance on AI can be substantial. Chatbot AI does not resolve this problem because its predictions are also restricted to the user's factor of interest.

These results show that the aggregation rule must reflect the information contained in individual and AI predictions. Clustering aggregation repeatedly counts the full prediction target when paired with Omniscient AI, whereas averaging aggregation attenuates factor-specific predictions when paired with Chatbot AI.

Supplementary Figure

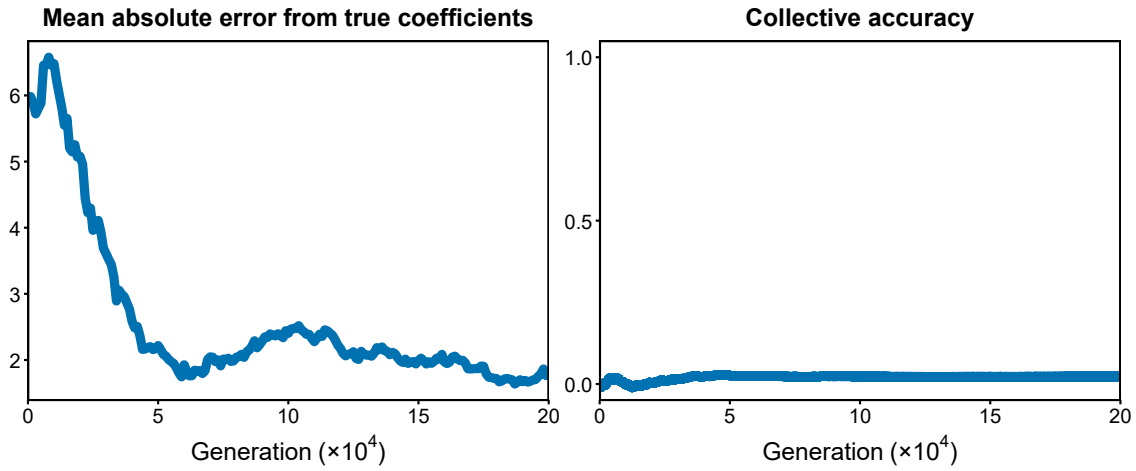


Figure S1: **Niche-expert structure with averaging aggregation cannot promote CI without AI.** Trajectories of the mean absolute error from true coefficients and collective accuracy under the niche-expert structure with averaging aggregation in the absence of AI. The mean absolute error from true coefficients is calculated by first taking, for each interest, the absolute difference between the true coefficient and the mean belief of individuals sharing that interest, and then averaging these differences across all $M+1$ interests. The mean absolute error decreases across generations, indicating that the mean beliefs move closer to their corresponding true coefficients. However, collective accuracy remains near zero, indicating that simply averaging individual predictions does not yield accurate collective predictions. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 100^2)$.

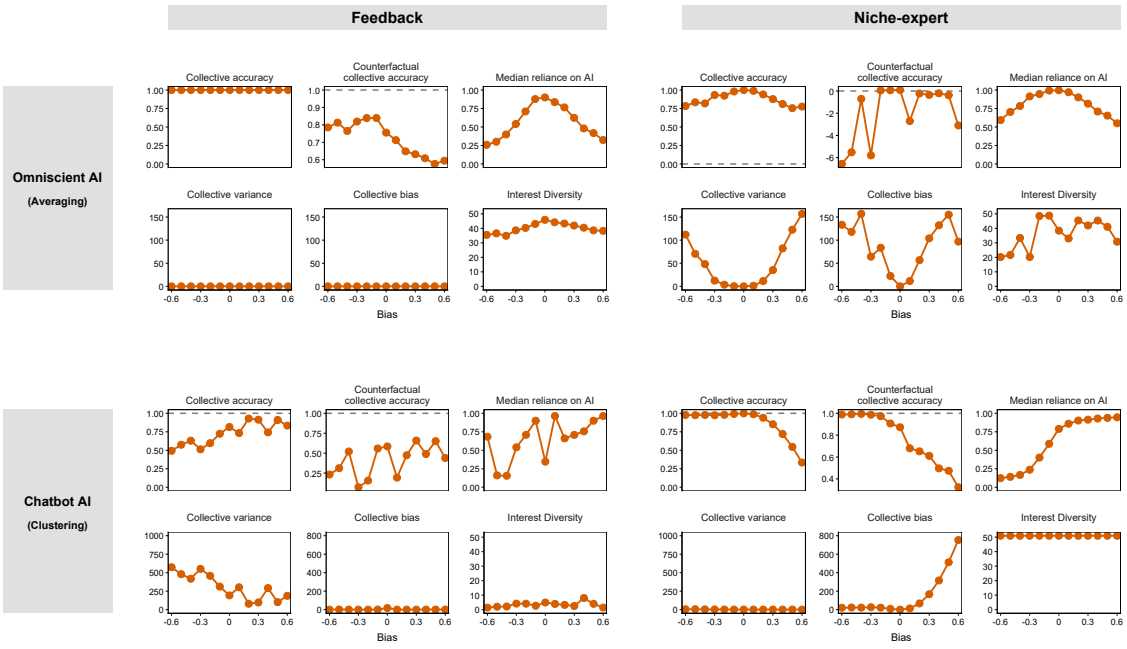


Figure S2: **Evaluation metrics across various levels of AI bias.** Evaluation metrics across varying levels of AI bias under different assumptions about AI types (Omniscient AI, and Chatbot AI) and incentive structures (feedback, niche-expert). Larger absolute AI bias indicates a less accurate AI model. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, and counterfactual human CI represents the group's predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as Hill-Shannon diversity of interest within the population, median reliance on AI represents the median value of reliance on AI (β). Collective bias represents systematic deviation of the collective prediction from the true target, collective variance represents variability of the collective prediction error across prediction events. All the evaluation metrics are averaged over generations 190,001–200,000. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$.

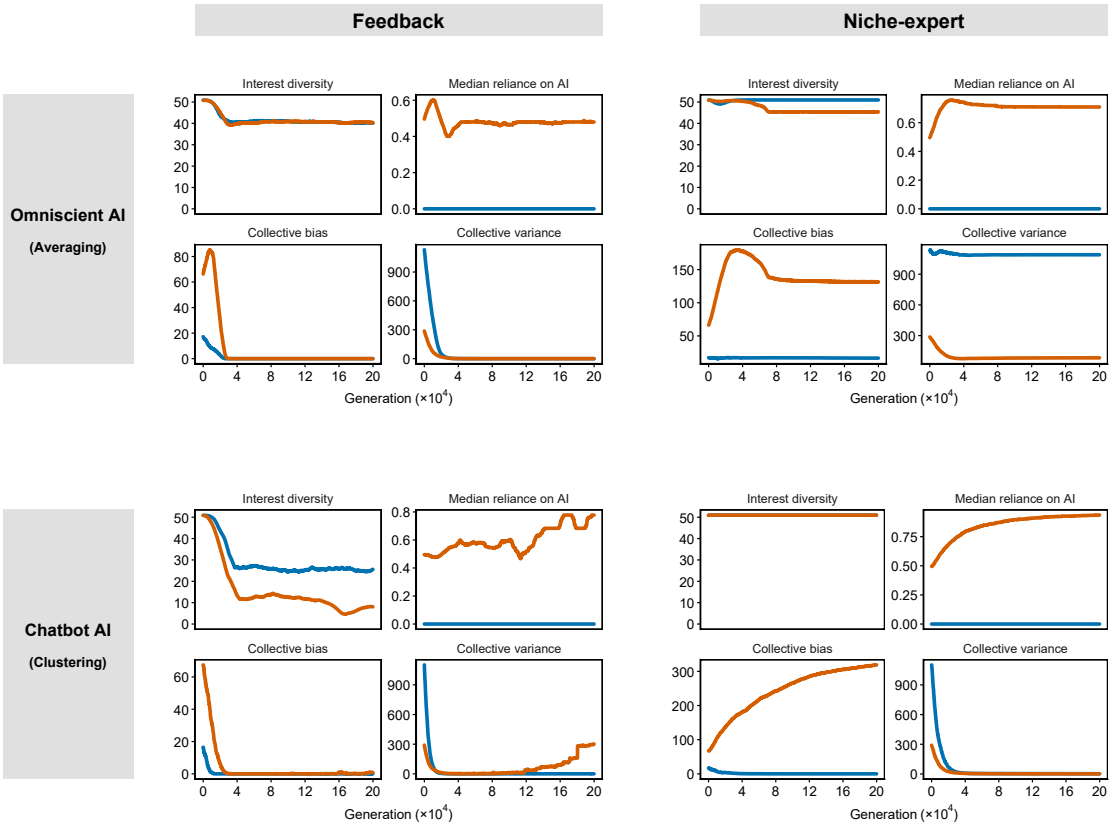


Figure S3: **Comparison of evaluation metrics with and without AI.** Trajectories of evaluation metrics when the group is assisted by the Omniscient AI and Chatbot AI. Here, feedback and niche-expert incentive structures are considered. Blue lines represent trajectories of evaluation metrics without AI assistance. Orange lines represent trajectories of evaluation metrics with AI assistance. Interest diversity is define as Hill-Shannon diversity of interest within population, median reliance on AI represent population median of reliance on AI (β), collective bias represent systematic deviation of the collective prediction from the true target, collective variance represent variability of the collective prediction error across prediction events. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

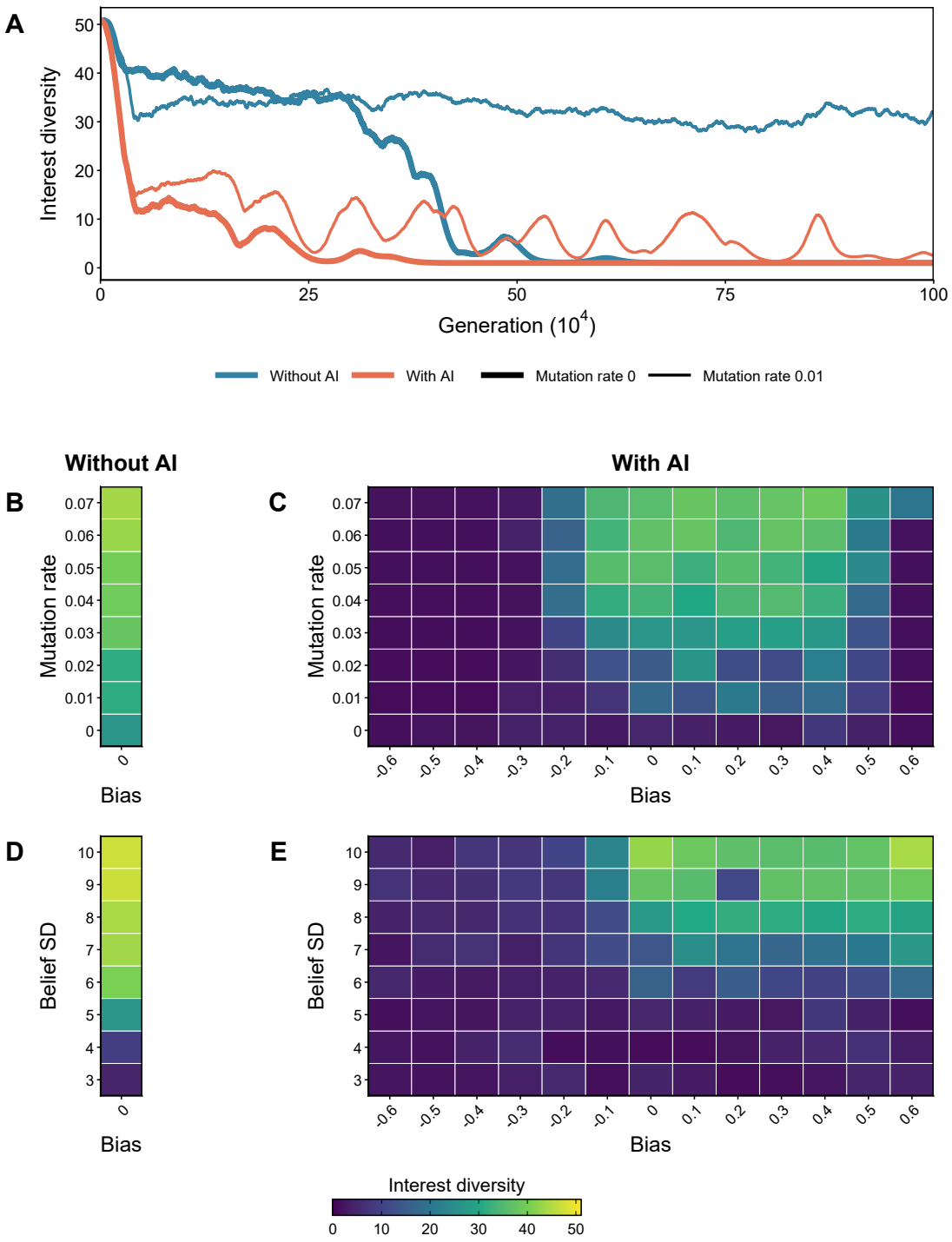

A
Interest diversity
Generation (10^4)
Without AI
With AI
Mutation rate 0
Mutation rate 0.01
Without AI
With AI
B
C
D
E
Mutation rate
Belief SD
Bias
Interest diversity

Figure S4: **Feedback structure becomes more prone to interest diversity loss with AI** Trajectories of interest diversity with and without AI under the feedback structure (A). Heatmaps of interest diversity across varying levels of mutation rate (B, C) and the standard deviation of the initial belief distribution (D, E), with and without AI. Interest diversity declines toward zero even without AI over long timescales when there is no mutation and the initial belief distribution is narrow. However, with AI, interest diversity declines more strongly, even under moderate mutation rates and wider initial belief distributions. Interest diversity in the heatmaps is averaged over generations 190,001–200,000. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$ for (D, E), $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 5^2)$ for (A, B, C), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$, Chatbot AI is used.

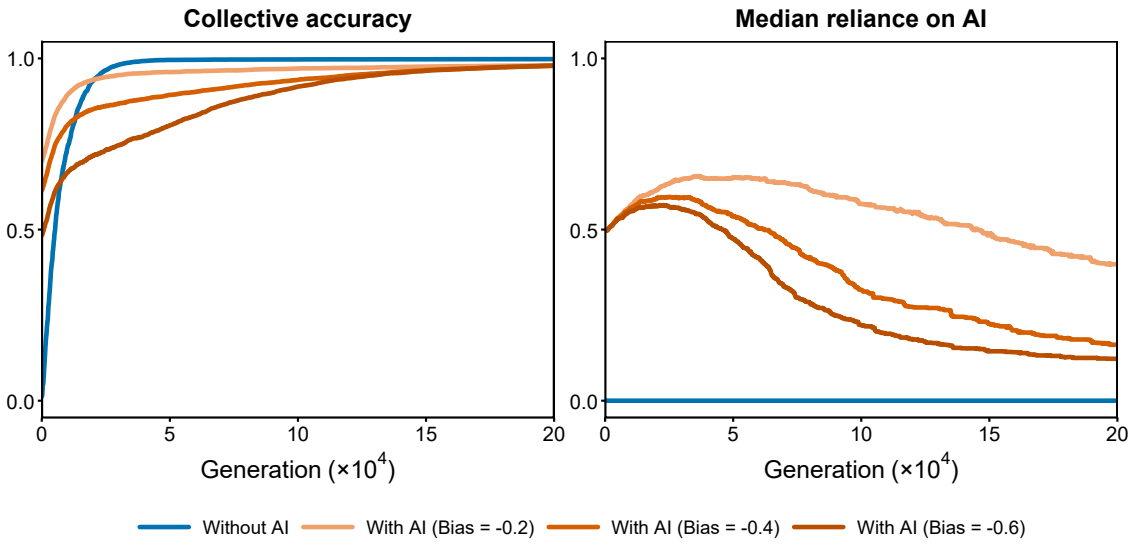


Figure S5: **Niche-expert structure may reduce reliance on AI, delaying the emergence of CI.** Trajectories of collective accuracy and median reliance on AI when the niche-expert structure causes reliance on AI to decline over time. Red lines represent trajectories with AI across varying levels of AI bias (-0.2, -0.4, and -0.6). Green lines represent trajectories without AI; reliance on AI is set to zero in the absence of AI. This reduction in reliance on AI occurs when the α_0 and AI bias have opposite signs, and is associated with the eventual emergence of CI, although CI emerges substantially later than in the condition without AI. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

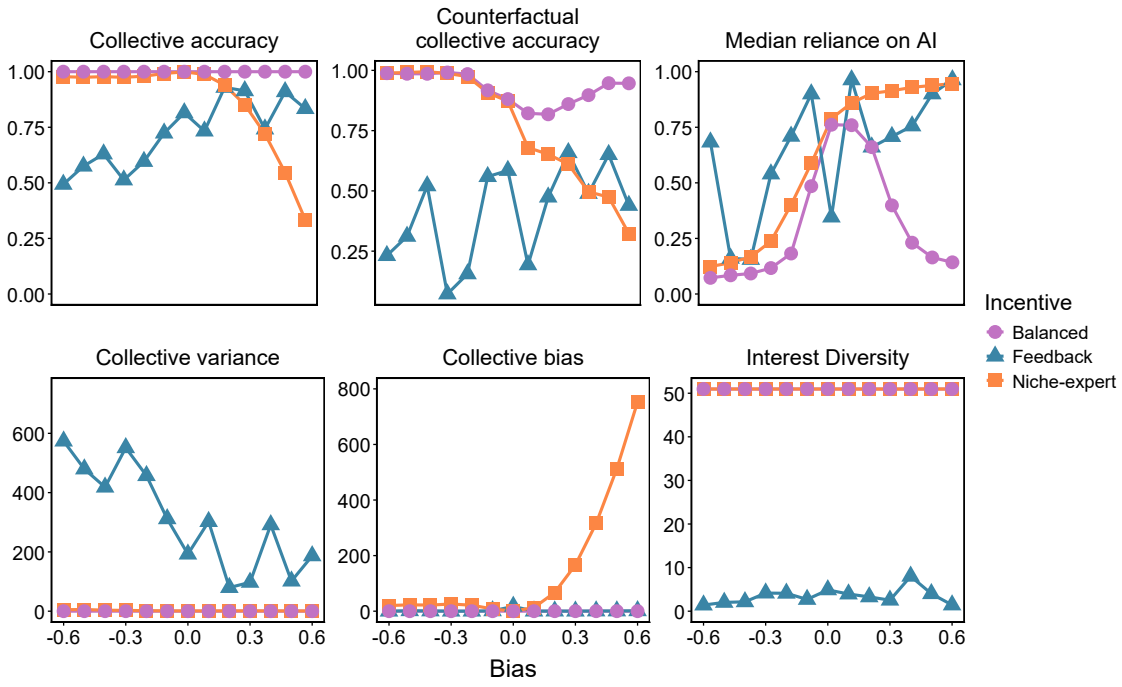


Figure S6: **Comparison of evaluation metrics between incentive structures across various levels of AI bias.** Evaluation metrics under feedback, niche-expert and balanced structures across varying levels of AI bias. Larger absolute AI bias indicates a less accurate AI model. All the evaluation metrics are averaged over generations 190,001–200,000. The balanced structure maintains near-perfect collective accuracy across bias levels and higher counterfactual human CI than the other incentive structures. These patterns are driven by well-calibrated reliance on AI, which increases as AI becomes more accurate, and by well-preserved interest diversity, which together lead to low collective variance and collective bias. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

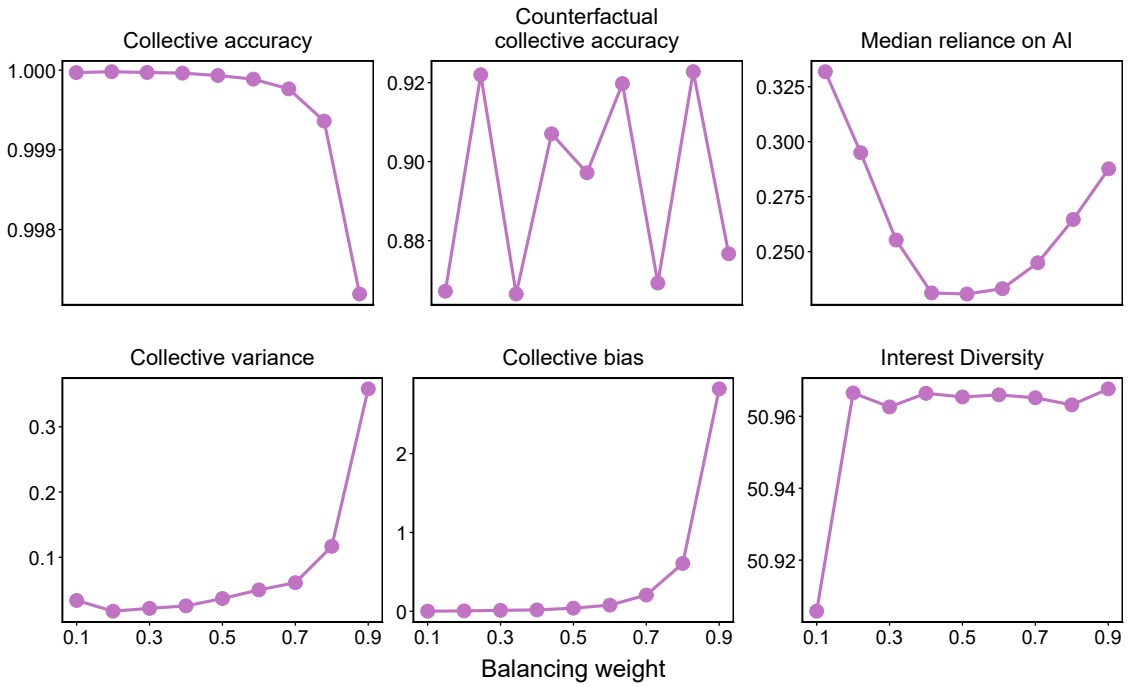


Figure S7: **Balanced structure is robust across varying balancing weights.** Evaluation metrics across varying levels of the balancing weight (0.1 – 0.9). The balancing weight represents the weight assigned to the niche-expert structure when constructing the balanced structure by combining the niche-expert and feedback structures. A balancing weight of 0.5 represents equal weighting of the niche-expert and feedback structures, whereas 0 corresponds to the feedback structure and 1 corresponds to the niche-expert structure. Although the evaluation metrics vary as the balancing weight changes from 0.1 to 0.9, the absolute magnitude of these deviations remains limited. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$, $\tau = 0.3$, Chatbot AI is used.

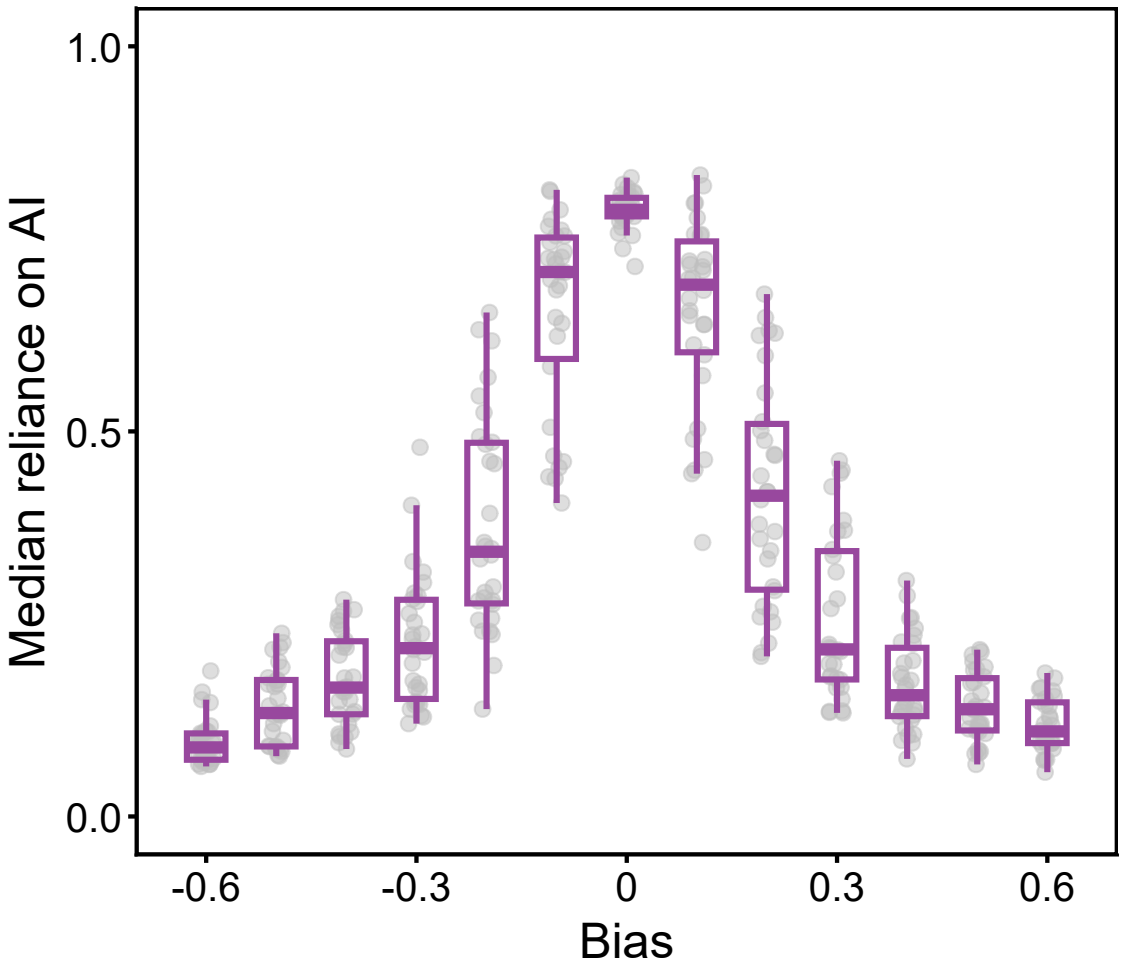


Figure S8: **Balanced structure shows well-calibrated reliance on AI.** Median reliance on AI across varying levels of AI bias. Thirty randomly initialized simulations were performed for each bias level. Each dot represents a single simulation, and the box plots show the median and interquartile range (IQR) for each bias level. The balanced structure shows well-calibrated reliance on AI, with reliance increasing as absolute magnitude of AI bias decreases. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

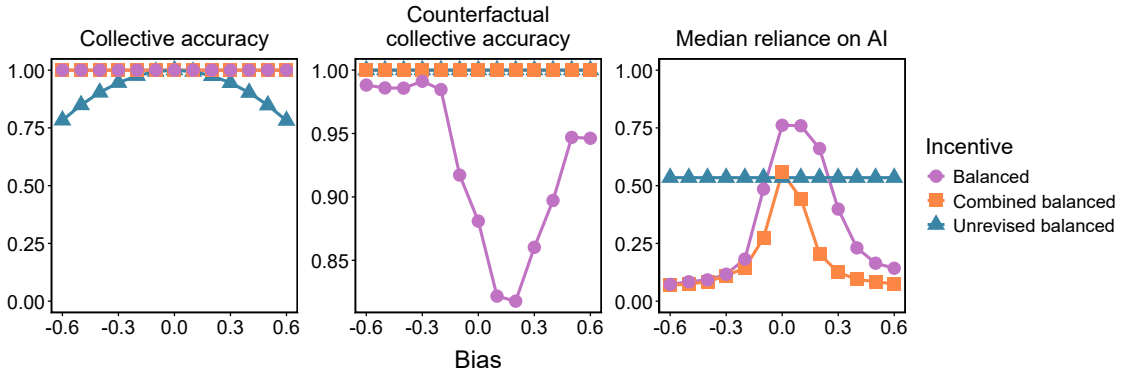


Figure S9: **Incentive structures incorporating unrevised prediction accuracy can promote better counterfactual human CI.** Evaluation metrics under the balanced, unrevised balanced, and combined balanced structures across varying levels of AI bias. The unrevised balanced structure is defined as the arithmetic mean of the feedback and niche-expert structures based on unrevised predictions, whereas the combined balanced structure is defined as the arithmetic mean of the balanced and unrevised balanced structures. All evaluation metrics are averaged over generations 190,001–200,000. The unrevised balanced structure achieves near-perfect counterfactual human CI; however, median reliance on AI remains near 0.5 because the unrevised balanced incentive does not account for AI predictions and therefore imposes no selection on β_A. Consequently, collective accuracy directly inherits AI bias and decreases as the magnitude of AI bias increases. The combined balanced structure, which leverages both the balanced and unrevised balanced structures, achieves near-perfect counterfactual human CI without compromising collective accuracy. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$; Chatbot AI is used.

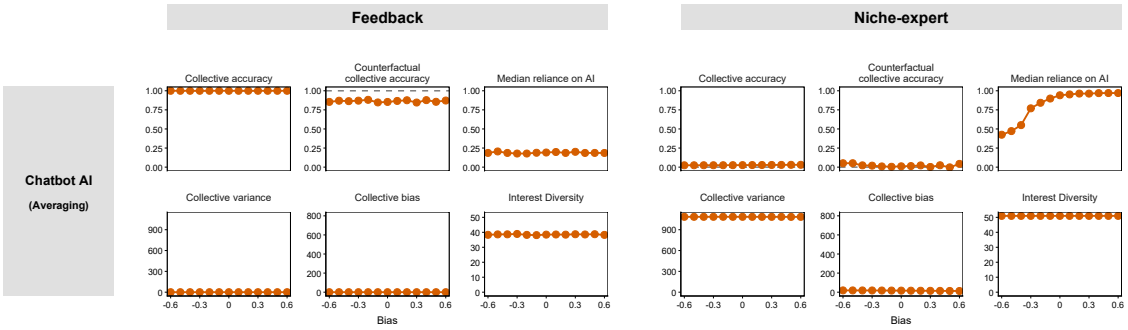


Figure S10: **Chatbot AI with averaging aggregation has a limited influence on CI.** Evaluation metrics across varying levels of AI bias under Chatbot AI with averaging aggregation. Two incentive structures, feedback and niche-expert, are considered. A larger absolute AI bias indicates a less accurate AI model. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, whereas counterfactual human CI represents the group's predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as the Hill–Shannon diversity of interests within the population, and median reliance on AI represents the median reliance on AI (β) across individuals. Collective bias represents the systematic deviation of the collective prediction from the true target, whereas collective variance represents the variability of collective prediction errors across prediction events. All evaluation metrics are averaged over generations 190,001–200,000. Prediction target parameters: $M = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 100^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$.

Supplementary Table

Table S1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\mathrm{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0.
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\mathrm{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\mathrm{Var}(Y - \hat{Y}^{\mathrm{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^{M} \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\mathrm{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

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